# Adam Robert Swietek

Postdoctoral Scholar | Department of Urban Planning and Spatial Analysis
USC Sol Price School of Public Policy | Urban Data Lab

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## ACADEMIC APPOINTMENT

**2025-present** **University of Southern California, Los Angeles**
Postdoctoral Scholar, Department of Urban Planning and Spatial Analysis, USC Price School; Urban Data Lab

## EDUCATION

**2024** **EPFL - PhD, Architecture and Sciences of the City**
Thesis: Spatial Modeling for Building Design Evaluation: From Visual Landscape Quality Assessment to Devaluation Risk Estimation
Advisor: Philippe Thalmann. Committee: Marilyne Andersen, Alexandre Alahi, Lint Barrage, and Silvia Tobias. Nominated for EPFL Thesis Distinction.

**2019** **Columbia University - MS, Real Estate Development and Finance**

**2013** **Georgia Institute of Technology - BS, Mechanical Engineering**
Minor in Energy Systems

## RESEARCH INTERESTS

Environmental externalities in cities; urban, real estate, and environmental economics; spatial modeling and GeoAI; urban data science; building performance; computational design; climate adaptation and resilience.

## PUBLICATIONS AND PREPRINTS

**Swietek, A. R.** (2026). "Structure-to-Structure Connectivity and Urban Wildfire Destruction Risk in California." SSRN preprint; under review at *Nature Cities*.

**Swietek, A. R.** (2024). "Automated Design Appraisal: Estimating Real Estate Price Growth and Value at Risk Due to Local Development." SSRN working paper.

**Swietek, A. R.**, & Zumwald, M. (2023). "Visual Capital: Evaluating Building-Level Visual Landscape Quality at Scale." *Landscape and Urban Planning*, 240, 104880.

**Swietek, A. R.**, & Schweizer, J. (2022). "How View Determines Well-Being and Value." Wuest Partner.

## SELECTED RESEARCH IN PROGRESS

- **Urban Electrification:** AI-based analysis of electrical permits in Los Angeles.
- **Hidden Housing:** Detecting unpermitted dwelling units in Los Angeles.
- **Construction and Demolition Waste:** Analysis of Facebook Marketplace listings.
- **MeshCity:** A Python toolkit for loading and transforming USGS LiDAR point clouds.
- **Generative Design for Reducing Urban Fire Spread and Structure Loss.**
- **Estimating Spatial Basis Risk to Hedge Extreme Temperatures with Prediction Markets.**
- **Urban Geometry of Heat Island Intensity.**
- **Openview:** Open-source visual-network and viewpoint-simulation tools for large-scale urban visibility analysis.

## GRANTS, FELLOWSHIPS, AND INNOVATION AWARDS

**2023** **AVIA - ENAC Innovation Seed Grant**
Principal investigator; integrated data-driven impact assessment for early-stage architectural design optimization.

**2022** **EPFL Blaze Trailblazer - REON**
Co-founder; automated reuse assessment for building components and construction materials.

**Circular Economy Transition Incubator - REON**
Selected for the nationwide incubator supporting circular-economy ventures.

**2019-2024** **Marie Sklodowska-Curie Fellowship - EU Horizon 2020**
European Union Horizon 2020 research and innovation programme, grant agreement No. 754354.

## TEACHING

**Economic Growth and Sustainability I (HUM-471)**
Teaching Assistant, EPFL Humanities and Social Sciences Program. Instructor: Philippe Thalmann.

**Economic Growth and Sustainability II (HUM-470)**
Teaching Assistant, EPFL Humanities and Social Sciences Program. Instructor: Philippe Thalmann.

## STUDENT SUPERVISION

**2024** **Esteban Requena - Master's thesis**
"Identifying Geographical Clusters of High or Low Sustainability Ratings and Their Implications."

**2023** **Yannick Neypatraiky - Master's thesis, EPFL Civil Engineering**
Data-Driven Assessment of the Swiss Building Stock: Characterising Building Typologies for Energy Retrofit Prioritisation.

**2022** **Paul Habert, Yannick Neypatraiky, and Thomas Rimbot - Machine Learning course project**
"Best Place for You to Live."

## SELECTED PRESENTATIONS AND TALKS

**2026** **"Openview: A Tool for Mesh-Based Urban Visibility Modeling and Analysis"**
2026 AAG Annual Meeting, San Francisco, California, March 17-21.

**2024** **Spatial Modeling in Building Design Evaluation**
WSL Landscape Centre Seminar Series, July 2.

**2023** **Spatial Statistics 2023: Climate and the Environment**
University of Colorado Boulder, July 18-21.

**2023** **"Visual Capital: Large-Scale Assessment of Building-Level Visual Landscape Quality"**
EPFL EDAR Doc.Days, April 17. Program | Video

**2020** **Forming Groups of Students Best Suited to Work Together on a Sustainability Issue**
With Philippe Thalmann. EPFL Teaching Sustainability Think Tank, November 23.